

The Mystery of the Heavy Little Cube



Zara and Eli were helping Ms. Romero set up the science table when they found two cubes. They looked exactly the same size. But when Zara lifted the silver cube, her eyes went wide.

‘This one is much heavier!’ she said.

‘How can two things that look the same feel so different?’ Eli asked.

Ms. Romero smiled. ‘Today,’ she said, ‘you are going to solve a mystery about mass, volume, and density.’



Topic: Mass, Volume, and Density

Volume: The Amount of Space



Next, Ms. Romero asked the class to look closely at the size of the cubes.

‘Volume is the amount of space something takes up,’ she said.

‘These two cubes take up the same amount of space, so they have the same volume.’

Zara grinned. ‘Same volume, different mass,’ she said.

‘We are getting closer to the answer.’



Topic: Mass, Volume, and Density

Mass: The Amount of Matter



Ms. Romero placed the cubes on a balance.

The silver cube pulled the scale down.

‘Mass tells us how much matter an object has,’ she explained. ‘Even if two objects look alike, one can have more matter packed inside.’

Zara touched the metal cube. ‘So this one has more mass,’ she said.

Eli nodded. ‘That is why it felt heavier!’



Topic: Mass, Volume, and Density

Science Element



Mass is how much matter an object has.

Volume is how much space an object takes up.

Density is mass per unit volume, or $\text{Density} = \text{Mass} \div \text{Volume}$.

Zara and Eli learned that two objects can be the same size but still have different masses.

The object with more mass packed into the same volume has greater density.



Floating and Sinking



Then the class tested other objects in water. A wooden block floated, but a rock sank to the bottom.

‘Density helps explain this too,’ Ms. Romero said.

‘Objects that are denser than water usually sink. Objects that are less dense than water can float.’

Zara watched the block bob at the top.

‘Density is everywhere!’



Topic: Mass, Volume, and Density

Density: The Big Clue



Ms. Romero wrote a new clue on the board: $\text{Density} = \text{Mass} \div \text{Volume}$.

‘Density tells how much mass is packed into each unit of space,’ she said.

‘Because the silver cube has more mass in the same volume, it is denser than the wooden cube.’

Eli smiled. ‘So density is the real mystery answer!’



Topic: Mass, Volume, and Density